Curcumin (diferuloylmethane) delivery methods: a review.

Curcumin interacts with a large number of extra- and intracellular targets in a biphasic dose-dependent manner. It controls inflammation, oxidative stress, cell survival, cell secretion, homeostasis, and proliferation. Its mechanisms of action are generally directed toward cells that exhibit disordered physiology or blatant mutation-based abnormal states. Optimizing preventative or therapeutic applications require delivering appropriate quantities of curcumin to lesioned cellular targets. Since diseased conditions anatomically are located from topical to systemic sites, efficient application of curcumin requires specific lesion-oriented delivery methods, representatives of which are here reviewed.